

Pluggable Database backup and restore activity

- Login the PDB database

SQL> conn

Enter user-name: sys/oracle@pdb1 as sysdba

Connected.

```
SQL> conn
Enter user-name: sys/oracle@pdb1 as sysdba
Connected.
SQL>
SQL>
SQL> show pdbs;
```

CON_ID	CON_NAME	OPEN MODE	RESTRICTED
3	PDB1	MOUNTED	

- After that we have to open the Pluggable Database.

SQL> alter pluggable database pdb1 open;

Pluggable database altered.

```
SQL> alter pluggable database pdb1 open;
Pluggable database altered.
SQL> show pdbs;
```

CON_ID	CON_NAME	OPEN MODE	RESTRICTED
3	PDB1	READ WRITE	NO

- Check the datafile under pdb1 database below the following command.

SQL> select name from v\$datafile;

```
SQL> select name from v$datafile;

NAME
-----
/u01/app/oracle/oradata/CDB/pdb1/system01.dbf
/u01/app/oracle/oradata/CDB/pdb1/sysaux01.dbf
/u01/app/oracle/oradata/CDB/pdb1/undotbs01.dbf
/u01/app/oracle/oradata/CDB/pdb1/users01.dbf
/u01/app/oracle/oradata/CDB/pdb1/sha01.dbf
```

- Now take a physical backup on the PDB.

[oracle@localhost ~]\$ rman target sys/oracle@pdb1

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connected to target database: CDB:PDB1 (DBID=2205859277)

```

RMAN> backup database plus archivelog;

Starting backup at 25-JUL-26
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=65 device type=DISK
skipping archived logs when connected to a PDB
backup cancelled because there are no files to backup
Finished backup at 25-JUL-26

Starting backup at 25-JUL-26
using channel ORA_DISK_1
channel ORA_DISK_1: starting full datafile backup set
channel ORA_DISK_1: specifying datafile(s) in backup set
input datafile file number=00010 name=/u01/app/oracle/oradata/CDB/pdb1/sysaux01.dbf
input datafile file number=00017 name=/u01/app/oracle/oradata/CDB/pdb1/sha01.dbf
input datafile file number=00009 name=/u01/app/oracle/oradata/CDB/pdb1/system01.dbf
input datafile file number=00011 name=/u01/app/oracle/oradata/CDB/pdb1/undotbs01.dbf
input datafile file number=00012 name=/u01/app/oracle/oradata/CDB/pdb1/users01.dbf
channel ORA_DISK_1: starting piece 1 at 25-JUL-26
channel ORA_DISK_1: finished piece 1 at 25-JUL-26
piece handle=/u01/app/oracle/fast recovery area/CDB/57583749491638EDE065000000000001/backupset/2026_07_25/01_mf_nnndf_TAG20260725T153803

```

After complete the backup check the backup list.

RMAN> list backup;

```

RMAN> list backup;

List of Backup Sets
=====

BS Key   Type LV Size       Device Type Elapsed Time Completion Time
-----
3        Full  491.33M   DISK        00:00:22    25-JUL-26
BP Key: 3  Status: AVAILABLE Compressed: NO Tag: TAG20260725T153803
Piece Name: /u01/app/oracle/fast_recovery_area/CDB/57583749491638EDE065000000000001/backupset/2026_07_25/01_mf_nn
9x_.bkp
List of Datafiles in backup set 3
File LV Type Ckp SCN    Ckp Time  Abs Fuz SCN Sparse Name
-----
9        Full  2346369   25-JUL-26                NO    /u01/app/oracle/oradata/CDB/pdb1/system01.dbf
10       Full  2346369   25-JUL-26                NO    /u01/app/oracle/oradata/CDB/pdb1/sysaux01.dbf
11       Full  2346369   25-JUL-26                NO    /u01/app/oracle/oradata/CDB/pdb1/undotbs01.dbf
12       Full  2346369   25-JUL-26                NO    /u01/app/oracle/oradata/CDB/pdb1/users01.dbf
17       Full  2346369   25-JUL-26                NO    /u01/app/oracle/oradata/CDB/pdb1/sha01.dbf

```

- Suppose Delete the sha01.dbf file from the existing location.

```

[oracle@localhost pdb1]$ pwd
/u01/app/oracle/oradata/CDB/pdb1
[oracle@localhost pdb1]$
[oracle@localhost pdb1]$
[oracle@localhost pdb1]$ ls -ltr
total 743000
-rw-r-----. 1 oracle oracle 37756928 Jul 25 10:36 temp01.dbf
-rw-r-----. 1 oracle oracle 5251072 Jul 25 15:38 users01.dbf
-rw-r-----. 1 oracle oracle 104865792 Jul 25 15:53 undotbs01.dbf
-rw-r-----. 1 oracle oracle 367009792 Jul 25 15:53 sysaux01.dbf
-rw-r-----. 1 oracle oracle 283123712 Jul 25 15:53 system01.dbf

```

- When I restart the database, the database is not restarted.

```
SQL> shu immediate
Pluggable Database closed.
SQL>
SQL>
SQL> show pdbs;

  CON_ID CON_NAME                                OPEN MODE  RESTRICTED
-----
      3 PDB1                                     MOUNTED

SQL>
SQL> alter pluggable database pdb1 open;
alter pluggable database pdb1 open
*
ERROR at line 1:
ORA-01157: cannot identify/lock data file 17 - see DBWR trace file
ORA-01110: data file 17: '/u01/app/oracle/oradata/CDB/pdb1/sha01.dbf'
```

- Now restore & recover the database

RMAN> restore database;

```
RMAN> restore database;

Starting restore at 25-JUL-26
using target database control file instead of recovery catalog
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=87 device type=DISK

channel ORA_DISK_1: starting datafile backup set restore
channel ORA_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_DISK_1: restoring datafile 00009 to /u01/app/oracle/oradata/CDB/pdb1/system01.dbf
channel ORA_DISK_1: restoring datafile 00010 to /u01/app/oracle/oradata/CDB/pdb1/sysaux01.dbf
channel ORA_DISK_1: restoring datafile 00011 to /u01/app/oracle/oradata/CDB/pdb1/undotbs01.dbf
channel ORA_DISK_1: restoring datafile 00012 to /u01/app/oracle/oradata/CDB/pdb1/users01.dbf
channel ORA_DISK_1: restoring datafile 00017 to /u01/app/oracle/oradata/CDB/pdb1/sha01.dbf
channel ORA_DISK_1: reading from backup piece /u01/app/oracle/fast_recovery_area/CDB/57583749491638EDE065000000000001/backupset/2026_07_2
_TAG20260725T153803_o692r49x_.bkp
```

RMAN> recover database;

```
RMAN> recover database;

Starting recover at 25-JUL-26
using channel ORA_DISK_1

starting media recovery
media recovery complete, elapsed time: 00:00:01

Finished recover at 25-JUL-26
```

- Open pluggable database

RMAN> alter pluggable database pdb1 open;

Statement processed

```
SQL> show pdbs;

  CON_ID CON_NAME                                OPEN MODE  RESTRICTED
-----
      3 PDB1                                     READ WRITE NO
```

- Check the datafile on OS level & SQL level.

```
SQL> select name from v$datafile;
```

```
NAME
```

```
-----  
/u01/app/oracle/oradata/CDB/pdb1/system01.dbf  
/u01/app/oracle/oradata/CDB/pdb1/sysaux01.dbf  
/u01/app/oracle/oradata/CDB/pdb1/undotbs01.dbf  
/u01/app/oracle/oradata/CDB/pdb1/users01.dbf  
/u01/app/oracle/oradata/CDB/pdb1/sha01.dbf
```

```
[oracle@localhost pdb1]$ ls -ltr
```

```
total 1050208
```

```
-rw-r-----. 1 oracle oracle 37756928 Jul 25 10:36 temp01.dbf  
-rw-rw----. 1 oracle oracle 314580992 Jul 25 16:01 sha01.dbf  
-rw-r-----. 1 oracle oracle 5251072 Jul 25 16:01 users01.dbf  
-rw-r-----. 1 oracle oracle 367009792 Jul 25 16:07 sysaux01.dbf  
-rw-r-----. 1 oracle oracle 283123712 Jul 25 16:07 system01.dbf  
-rw-r-----. 1 oracle oracle 104865792 Jul 25 16:07 undotbs01.dbf
```